



IPB University
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Kecerdasan Buatan **Kuliah 7 - First Order Logic**

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Outline

- Pembahasan soal kuis minggu lalu
- Review Propositional Logic
- Konsep dasar First Order Logic (FOL)
- Syntax dan Semantics dalam FOL
- Latihan



Review materi minggu lalu

- Logical agents apply **inference** to a **knowledge base** to derive new information and make decisions
- Basic concepts of logic:
 - **syntax**: formal structure of **sentences**
 - **semantics**: **truth** of sentences wrt **models**
 - **entailment**: necessary truth of one sentence given another
 - **inference**: deriving sentences from other sentences
 - **soundness**: derivations produce only entailed sentences
 - **completeness**: derivations can produce all entailed sentences
- Propositional logic lacks expressive power



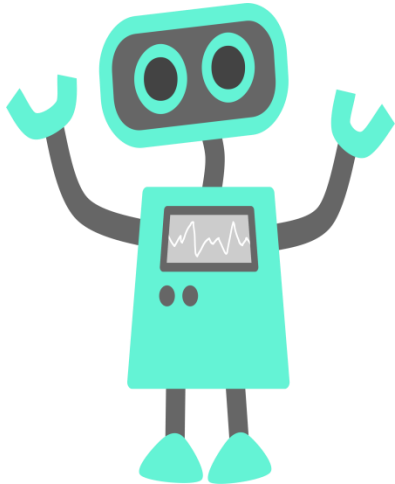
Review Propositional Logic



- ❑ **Logical constants:** true, false
- ❑ **Propositional symbols:** $P, Q, S, \dots$ (**atomic sentences**)
- ❑ **Wrapping parentheses:** $(\dots)$
- ❑ Sentences are combined by **connectives**:
 - $\wedge$...and [conjunction]
 - $\vee$...or [disjunction]
 - $\Rightarrow$...implies [implication / conditional]
 - $\Leftrightarrow$..is equivalent [biconditional]
 - $\neg$...not [negation]
- ❑ **Literal:** atomic sentence or negated atomic sentence



Konsep Dasar First Order Logic (FOL)



- **Propositional logic** only deals with “facts”, statements that may or may not be true of the world, e.g. “It is raining”.
- **But**, one cannot have variables that stand for books or tables.
- In **first-order logic**, variables refer to things in the world and, furthermore, you can quantify over them – to talk about all of them or some of them without having to name them explicitly.



FOL Syntax

Term

- Constant symbols: Fred, Japan, Bacterium39
- Variables: x , y , a
- Function symbol: $F(x)$, Mother-of(John)

Sentence

- A predicate symbol applied to zero or more terms: $\text{on}(a, b)$, $\text{sister}(\text{Jane}, \text{Joan})$, $\text{sister}(\text{Mother-of}(\text{John}), \text{Jane})$, $\text{its-raining}()$.
- $t_1 = t_2$.
- For v a variable and Φ a sentence, then $\forall v. \Phi$ and $\exists v. \Phi$ are sentences.

"for all""there exists"
- Syntax of Propositional Logic: $\vee$, $\wedge$, $\Rightarrow$, $\Leftrightarrow$, $\neg$



Sintaks FOL : Kalimat Atomik

- Definisi atomic sentence :
predicate($term_1, \dots, term_n$)
atau $term_1 = term_2$
function($term_1, \dots, term_n$)
atau constant
atau variable
- Contoh :
 - **Brother** (KingJohn, RichardTheLionheart)





Sintaks FOL : Kalimat Kompleks

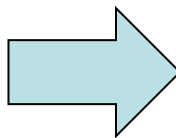
Kalimat kompleks: kalimat atomik yang digabungkan dengan *connectives*.

- Definisi complex sentence :

$$\neg S, \quad S_1 \wedge S_2, \quad S_1 \vee S_2, \quad S_1 \Rightarrow S_2, \quad S_1 \Leftrightarrow S_2$$

- Contoh :

- $\text{Sibling}(\text{KingJohn}, \text{Richard}) \Rightarrow \text{Sibling}(\text{Richard}, \text{KingJohn})$
- $>(1, 2) \vee \leq(1, 2)$
- $>(1, 2) \wedge \neg >(1, 2)$
- $\text{Belajar}(x, \text{SC}) \Rightarrow \text{Mengerti}(x, \text{AI})$



Semantics? benar/salah



Sintaks FOL : Universal Quantification

- ❑ **Syntax:**
Jika **S** kalimat, $\forall$ variables **S** adalah kalimat
- ❑ **Contoh:**
 - “Semua mahasiswa IPB adalah Genius”
 - $\forall x \text{ mahasiswa}(x, \text{IPB}) \Rightarrow \text{Genius}(x)$



Sintaks FOL : Universal Quantification

- ❑ Biasanya, $\Rightarrow$ adalah operator /connective yang digunakan dengan $\forall$.
- ❑ Masalah yang sering terjadi : menggunakan $\wedge$ sebagai connective untuk $\forall$:
 $\forall x \text{ mahasiswa}(x, \text{IPB}) \wedge \text{Genius}(x)$
- ❑ Kalimat ini berarti “Semua orang adalah mahasiswa IPB dan Genius”.



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Sintaks FOL : Existential Quantification

- ❑ **Syntax :**
Jika **S** kalimat, $\exists$ **variable S** adalah kalimat
- ❑ **Contoh:**
 - “Ada mahasiswa IPB yang pintar”
 - $\exists x \text{ mahasiswa}(x, \text{IPB}) \wedge \text{pintar}(x)$



Sintaks FOL : Existential Quantification

- ❑ Biasanya, $\wedge$ adalah operator /connective yang digunakan dengan $\exists$.
- ❑ Masalah yang sering terjadi : menggunakan $\Rightarrow$ sebagai connective untuk $\exists$:
 $\exists x \text{ mahasiswa}(x, \text{IPB}) \Rightarrow \text{pintar}(x)$
- ❑ Kalimat ini true jika ada setidaknya 1 orang (object) yang tidak kuliah di IPB



- $\forall x \forall y S$ sama dengan $\forall y \forall x S$, biasa ditulis $\forall x, y S$
- $\exists x \exists y S$ sama dengan $\exists y \exists x S$, biasa ditulis $\exists x, y S$
- $\exists x \forall y S$ TIDAK sama dengan $\forall y \exists x S$!
 - $\exists x \forall y \text{ Mencintai } (x, y)$
“Ada (sekurang-kurangnya) seseorang yang mencintai semua orang di dunia.”
 - $\forall y \exists x \text{ Mencintai } (y, x)$
“Semua orang di dunia mencintai sekurang-kurangnya satu orang”.





PRACTICE



- ❑ “Ayah adalah orangtua”
 $\forall x, y \text{ Ayah}(x, y) \Rightarrow \text{Orangtua}(x, y)$
- ❑ “Hubungan saudara berlaku simetris”
 $\forall x, y \text{ Saudara}(x, y) \Leftrightarrow \text{Saudara}(y, x)$
- ❑ “Ibu adalah orangtua berjenis kelamin perempuan”
 $\forall x, y \text{ Ibu}(x, y) \Leftrightarrow \text{Orangtua}(x, y) \wedge \text{Perempuan}(x)$
- ❑ “Sepupu adalah anak dari saudara orangtua”
 $\forall x, y \text{ Sepupu}(x, y) \Leftrightarrow \exists ox, oy \text{ Orangtua}(ox, x) \wedge \text{Saudara}(ox, oy) \wedge \text{Orangtua}(oy, y)$



Summary

- First order logic (FOL) lebih “luwes” / ekspresif dibandingkan dengan propositional logic.
- Dalam FOL, terdapat objects dan relations.
- Sama seperti propositional logic, FOL juga mempunyai syntax dan semantics.
 - **syntax**: formal structure of **sentences**
 - **semantics**: **truth** of sentences in **models**





Soal Quiz minggu ini – untuk didiskusikan

- $\exists x. \text{Big}(x) \wedge \text{Cat}(x)$
- $\exists x. \text{Big}(x) \Rightarrow \text{Cat}(x)$
- $\exists x. \text{Big}(x) \vee \text{Cat}(x)$
- $\forall x. \text{Big}(x) \wedge \text{Cat}(x)$
- $\forall x. \text{Big}(x) \Rightarrow \text{Cat}(x)$
- $\forall x. \text{Big}(x) \vee \text{Cat}(x)$



- A. All cats are big
- B. There is a small thing or a cat
- C. All big things are cats
- D. Everything is a big cat
- E. All cats are small
- F. There is a big cat
- G. All small things are cat
- H. There is a big thing or a cat

Ingat Logical equivalence!

- Two sentences are **logically equivalent** iff true in same models: $\alpha \equiv \beta$
iff $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$